

Dhanush Pujari

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EDUCATION

Mahindra University

Bachelor of Technology (B.Tech) in Artificial Intelligence

Graduated: Aug 2025

EXPERIENCE

Founder | VR Engineer

Aug 2025 – Present

OTR Technologies Pvt. Ltd.

- Founded OTR Technologies PVT. Ltd. And led the development of immersive Virtual Reality (VR) solutions for villa communities and residential apartment projects, enabling interactive property walkthroughs before construction completion.
- Designed and developed real-time VR experience using Unreal Engine 5, Blueprints, OpenXR, and Meta Quest, integrating architectural models from Sketchup and Blender, with photorealistic rendering in D5 Render.
- Collaborated with architects, builders to transform architectural designs into interactive VR experiences, improving client visualization and decision-making.

AR/VR Developer Intern

March 2025 – Oct 2025

Snapchat

- Contributed to the development of an indoor navigation API for Snapchat Spectacles using Augmented Reality.
- Assisted in improving spatial mapping algorithms and user experience for real-time navigation.
- Conducted testing and validation of AR navigation features to improve performance and accuracy.

Machine Learning Intern

Sept 2023 – Nov 2023

Corizo

- Performed data analysis on historical stock market datasets and trained baseline machine learning models.
- Developed classification models for wine quality prediction and evaluation.

App Developer

April 2021 – Aug 2021

HNWI Food Mart

- Designed and developed a mobile application based on business and user requirements.
- Conducted UX research, user testing, and UI/UX design improvements.
- Collaborated on feature planning and application optimization.

PROJECTS

AI Semantic Media Search Engine (Vector Embeddings & RAG) | Tech Stack: Python, PyTorch, Hugging Face, Qdrant, Streamlit

- Built a semantic search engine using vector embeddings and Qdrant to retrieve media through natural language queries.
- Optimized data ingestion pipelines, achieving 216+ records/second while preventing memory bottlenecks.
- Resolved concurrency and memory issues between Streamlit and PyTorch deployments.

AutoClip-ML: GPU-Accelerated Video Highlight Pipeline | Tech Stack: PyTorch, OpenAI Whisper, DistilBERT, FFmpeg, Pandas

- Developed an NLP-driven pipeline that automatically extracts high-retention video highlights.
- Managed GPU memory constraints for efficient local execution.
- Generated vertical short-form content with automated subtitles using FFmpeg.

End-to-End Algorithmic Trading Signal Predictor | Tech Stack: XGBoost, Pandas, yFinance, Scikit-Learn, FastAPI, React

- Developed a quantitative prediction platform for Nifty 50 market forecasting.
- Engineered financial indicators and addressed class imbalance challenges.
- Achieved 53.54% out-of-sample directional accuracy and a 24.77% cumulative back tested return.

AI Architectural Style Studio | Tech Stack: Stable Diffusion, ControlNet, OpenCV, FastAPI, React Three Fiber

- Built a generative AI platform for transforming architectural models into photorealistic renders.
- Leveraged ControlNet and OpenCV to preserve structural accuracy during image generation.
- Optimized inference pipelines for consumer-grade GPU hardware.

BodySync: Real-Time Virtual Try-On Application

- Developing a virtual try-on platform using computer vision and generative AI techniques for realistic clothing visualization.

AI-Powered Emotional Support Chatbot

- Developed a real-time emotional support chatbot using OpenAI Realtime APIs with speech-to-speech interaction and sentiment-aware responses.

TECHNICAL SKILLS

Machine Learning & AI: PyTorch, XGBoost, Hugging Face Transformers, NLP, RAG, Qdrant Vector Database

Programming Languages: Python, JavaScript, SQL

Data & Computer Vision: Pandas, NumPy, OpenCV, FFmpeg, Data Preprocessing, Feature Engineering

Web & Backend: FastAPI, Node.js, Express.js, React, React Three Fiber, Streamlit, MongoDB

LEADERSHIP & ACHIEVEMENTS

Head, 8Counts Dance Club – Mahindra University (2023 – 2024): Led university dance club activities, event planning, and team coordination. Awards: Top Participant – HULT Prize; Winner – Battle of the Beats; 1st Place – Francophone Competition
Interests: Artificial Intelligence, AR/VR, Dance, Calisthenics

CERTIFICATES

- **Google Cloud Big Data and Machine Learning Fundamental's:** Gained hands-on knowledge of big data processing, machine learning workflows, and cloud-based AI solutions using Google Cloud Platform (GCP).
- **Python Programming Course (Technophilia – IIT Kharagpur):** Learned Python programming fundamentals, object-oriented programming, data structures, and problem-solving techniques for software and machine learning applications.
- **Web Development Course (Technophilia – IIT Kharagpur):** Learned to build responsive web applications using HTML, CSS, JavaScript, and modern web development practices for front-end and back-end integration.
- **Android App Development Course (Technophilia – IIT Kharagpur):** Developed Android applications using Java/Android Studio, focusing on UI design, activity lifecycle, and mobile application development fundamentals